

Detecting the Malaria Parasite in Blood Smears with Transfer Learning and k -Nearest Neighbours

Working manuscript template — adapt it to your own solution

First Last

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Abstract

Malaria kills roughly 600,000 people every year, mostly where trained pathologists able to recognise the parasite under a microscope are scarce. We study whether this task can be automated: on a public dataset of 27 558 microscopy images of individual red blood cells (National Institutes of Health, NIH) we distinguish healthy cells from cells infected by the *Plasmodium* parasite. Our current solution first uses a frozen ResNet50 convolutional neural network as a feature extractor and then classifies the resulting vectors with a simple k -nearest-neighbours (k -NN) rule. This baseline reaches an area under the ROC curve of 0.93 on the test set. At the same time we show why accuracy alone is misleading in medicine: at the default decision threshold the model attains an accuracy of 0.77 but a sensitivity of only 0.56, missing nearly half of the infected cells. This document is a starting point; you replace it with your own, stronger classifier at the marked places.

1 Introduction and Problem Statement

Malaria remains one of the deadliest infectious diseases: according to the World Health Organization (WHO), about 600,000 people died of it in 2022 [1]. Reliable diagnosis still relies on the so-called thick drop and thin blood smear, where a technician searches red blood cells under a microscope for the *Plasmodium* parasite. This procedure is slow, requires an experienced specialist, and precisely in the regions with the highest malaria burden such specialists are critically scarce. This raises the question we address: *can the decision “is this cell infected?” be reliably delegated to an algorithm?*

The key obstacle for classical machine learning is that training an image recogniser from scratch requires millions of labelled images and substantial computing power. We sidestep both through *transfer learning*: we take a network pre-trained on millions of ordinary photographs, reuse its ability to “see” shapes and textures, and only re-train a small decision part. Concretely, in this draft we combine a frozen ResNet50 feature extractor with a k -nearest-neighbours classifier and reach an area under the ROC curve of 0.93 on the NIH dataset; at the same time we stress that the quality of a medical classifier must be judged by its sensitivity and specificity, not by accuracy alone.

This document is a **working template**. It describes the baseline solution and its evaluation so that you can start from it and replace it with your own, improved approach. The places where you should do so are marked “For the team”.

For the team: add one sentence here stating how your solution differs from the baseline and what main result it achieves.

2 Data

We use the public *Malaria Cell Images* dataset from the U.S. National Institutes of Health [2]. It contains 27 558 colour images of individual, already segmented red blood cells and is perfectly *balanced* — exactly half of the cells are infected and half are healthy (13,779 and 13,779). The balance matters for interpreting the results: it means that random guessing yields 50 % accuracy, so any improvement above this line comes from a genuinely learned distinction.

To make the results an honest measure of generalisation to new cells, we split the data into three parts in a *stratified* way (preserving the class ratio) and with a fixed random seed: a training set (19 289 cells), a validation set (4 135 cells) for tuning and model selection, and a held-out test set (4 134 cells) used exclusively for the final evaluation. A separate test set is essential: without it we could easily tune the model to the specific data and overestimate its true quality.

3 Method

3.1 Feature extraction with a frozen ResNet50

Because we have neither enough data nor time to train a convolutional neural network (CNN) from scratch, we use the ResNet50 network [3] pre-trained on the ImageNet dataset as a *feature extractor*. We first apply standard preprocessing to each cell image (resizing to 224×224 pixels and colour normalisation as expected by ImageNet) and then pass it through the network with its final classification layer removed. The output for each cell is a vector of **2048 numbers** — the *features*, which summarise what is in the image (shapes, textures, spots). We keep the network *frozen*: its weights do not learn, it merely translates an image into numbers. This reduces the whole task to classifying vectors, which even a weak computer handles in a few seconds.

3.2 Classification with k -nearest neighbours

As the initial classifier we deliberately chose the simplest possible model in order to obtain an honest *baseline* for comparison: the k -nearest-neighbours (k -NN) rule [4]. It does not actually train — it merely memorises the training features. It classifies a new cell by finding the k closest (most similar in feature space) among the memorised cells and letting them vote, with nearer neighbours weighted more heavily; the fraction of votes for the class “infected” is taken as the estimated probability. We used $k = 31$ neighbours and standardised the features before computing distances (zero mean, unit variance), because distances are only meaningful when the individual features share a comparable scale.

For the team: this is where you replace Section 3.2 with your own solution — for example a small trained neural network. Describe its architecture and how you tuned it.

4 Evaluation

4.1 Why accuracy alone is not enough

In medicine the two kinds of error carry very different costs, which is why we do not measure a classifier’s quality by *accuracy* (the fraction of correctly classified cells) alone. Let us denote as true positive (TP) an infected cell correctly flagged, false negative (FN) an infected cell wrongly declared healthy, false positive (FP) a healthy cell wrongly flagged as infected, and true negative

Table 1: Results on the held-out test set (4 134 cells). Accuracy, sensitivity and specificity are reported at the default threshold 0.5; the competition metric is the specificity at a sensitivity of ≥ 99 %. Values are from a single deterministic run.

Method	Accuracy	Sensitivity	Specificity	Spec. @sens. 99 %	AUC
ResNet50 + k -NN (baseline)	0.77	0.56	0.99	0.00	0.93
Your solution	—	—	—	—	—

(TN) a healthy cell correctly passed. From these we define two key quantities:

$$\text{sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \quad (1)$$

Sensitivity says what fraction of the truly sick the model catches; low sensitivity means missed patients, which can cost a life. Specificity says what fraction of the healthy the model correctly passes; low specificity means needless false alarms. These two quantities trade off against each other depending on the chosen *threshold* above which a cell is declared infected; the whole trade-off is captured by the ROC curve (receiver operating characteristic) and summarised by the area under it (area under the curve, AUC).

As the main **competition metric** we chose the *specificity at a sensitivity of ≥ 99 %*. For a deadly disease the non-negotiable requirement is “not missing a sick patient”, so we fix sensitivity high (catch at least 99 % of the infected) and compete on who keeps the highest specificity, that is the fewest false alarms. This corresponds to the point on the ROC curve at a sensitivity of 99 %.

4.2 Protocol

We “train” the classifier (for k -NN, merely memorise) on the training set, perform any threshold selection on the validation set, and measure all numbers reported below on the held-out test set (4 134 cells), which the model never sees during learning. The k -NN method is deterministic, so we report the results of a single run.

4.3 Results

The baseline results are summarised in [Table 1](#). The most telling point is the conflict between accuracy and sensitivity: at the default threshold of 0.5 the k -NN reaches an accuracy of 0.77 but a sensitivity of only 0.56 — as the confusion matrix in [Fig. 1](#) shows, it misses 904 of the 2067 infected cells. The high specificity (0.99) thus masks the fact that the model passes nearly half of the sick; accuracy alone would have deceived us. The full ROC curve ([Fig. 2](#)) shows that the *ranking* ability is not bad: the area under it is 0.93. Our competition metric is stricter, however. We require catching at least 99 % of the sick, and under this constraint the specificity of k -NN drops to 0.00 — to catch 99 % of the infected it would have to push the threshold so low that it flags nearly every cell. The simple baseline therefore fails to meet the clinical bar; this is exactly why a learned model is needed (your task).

5 Discussion

The weaker k -NN result is not an accident but a consequence of two effects worth noting. First, in the 2048-dimensional feature space distances stop being informative — all points become roughly

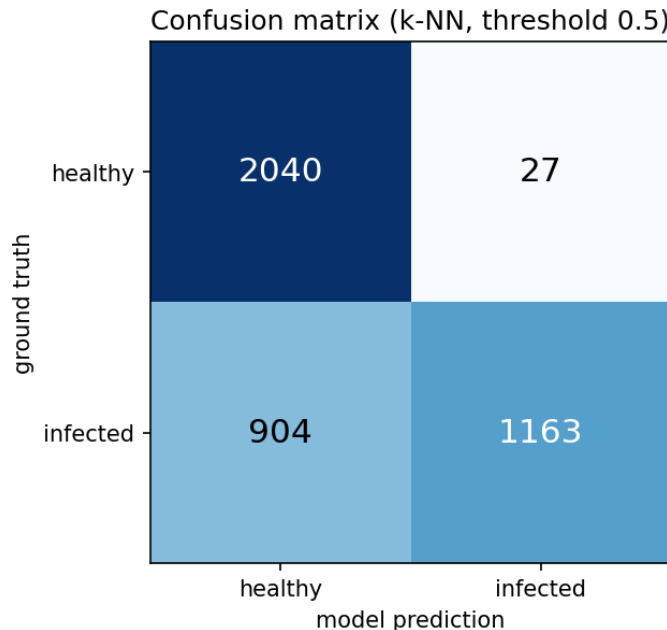


Figure 1: Confusion matrix of the k -NN baseline on the test set at threshold 0.5. Rows are the true classes, columns the predictions. The bottom-right cell (1163) are correctly caught sick cells, the bottom-left cell (904) are *missed* sick cells — the most expensive type of error.

equidistant, so the notion of a “nearest neighbour” loses its sharpness (the *curse of dimensionality*). Second, k -NN learns nothing: it cannot weigh which of the 2048 features matter for telling the parasite apart, treating them all equally. This is exactly where a better-informed model that learns what matters has room to improve.

Independently of the model used, two practical limitations remain. A real deployment would face *domain shift* — a different microscope, different staining, or a different patient population could lower the accuracy relative to our dataset. And because the cost of a missed sick patient exceeds the cost of a false alarm, in practice the model is better suited as a tool that *flags* cases for a clinician than as a final arbiter.

For the team: this is where the analysis of your results belongs — what worked, what did not, by how much you beat the baseline, and how you explain it.

6 Conclusion

We have shown that a frozen ResNet50 used as a feature extractor, combined with a simple k -nearest-neighbours classifier, reaches an area under the ROC curve of 0.93 on the NIH dataset, and we have stressed why medicine requires judging sensitivity and specificity rather than accuracy alone. This baseline is the starting point your solution is meant to beat — in particular by keeping the specificity as high as possible at the required sensitivity of 99 %.

References

- [1] World Health Organization. World Malaria Report 2023, 2023.

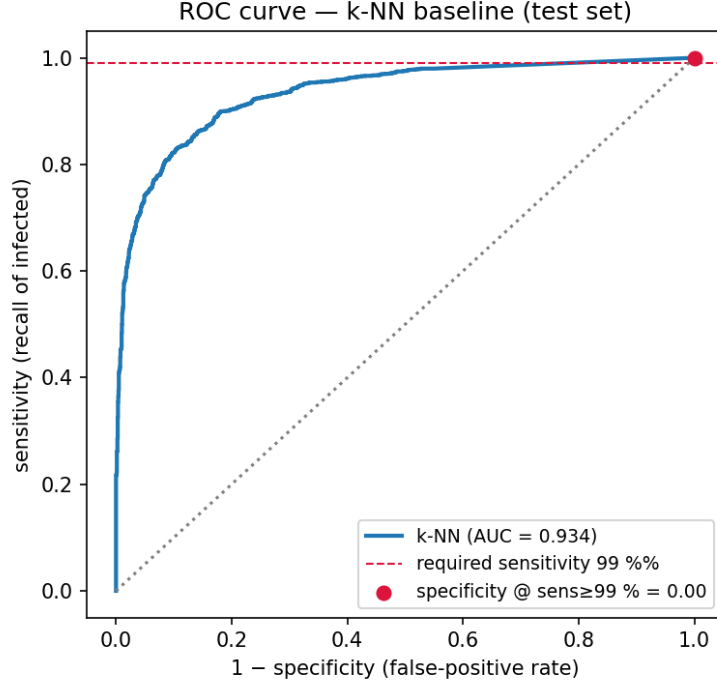


Figure 2: ROC curve of the k -NN baseline on the test set. The horizontal line marks the required sensitivity of 99 %; the competition metric is the highest specificity attainable above that line. The baseline only reaches it at a near-100 % false-positive rate (specificity ≈ 0).

- [2] Sivaramakrishnan Rajaraman, Sameer K. Antani, Mahdiah Poostchi, Kamolrat Silamut, Md A. Hossain, Richard J. Maude, Stefan Jaeger, and George R. Thoma. Pre-trained convolutional neural networks as feature extractors toward improved malaria parasite detection in thin blood smear images. *PeerJ*, 6:e4568, 2018.
- [3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [4] Thomas M. Cover and Peter E. Hart. Nearest Neighbor Pattern Classification. *IEEE Transactions on Information Theory*, 13(1):21–27, 1967.